

Key considerations for developing a Dash application

1. Please ensure you have added the correct column values as indicated in the hint for each code snippet. Example for **X and Y** in the charts
2. Please ensure that you have used the correct **indentation** in the code.
3. Fill in your code using the **placeholders(...)** provided in the code skeleton provided in the lab. Separate each component by commas.

Example: This creates a **Div element in HTML**, a container for other components. Inside this div, various child elements are placed, which need to be separated by commas.

```
html.Div([          html.Label("Select Statistics:"),          dcc.Dropdown(          i
d='.....',          options=.....,          value='..
.....',          placeholder='.....'          )
]),
```



Information:

Pressing Tab in the current editor will insert the tab character. Toggle this behavior by pressing Control+M.



Common errors:

1. Syntax error: invalid syntax def update_input_container(selected_statistics)

Solution: Check if any parentheses or square brackets are unclosed before the **def** function.

The HTML components are organized in a hierarchy using brackets. The use of square brackets **[]** **within html.Div()** or any other Dash component is essential for defining children elements in a list. This list-based structure allows you to include multiple child components inside a single-parent component.

Outer HTML

```
html.Div([ ...          ...          ])
```

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This `html.Div` is the container for the entire layout of the dashboard. It encapsulates and structures all the other HTML elements as a cohesive unit.

They use square brackets to group multiple child components within a single-parent component. This enables you to structure complex layouts with nested elements.

Dash components need to maintain a clear hierarchy to render correctly. The square brackets help maintain this hierarchy by clearly indicating which components are children of a particular parent component.

Inner HTML

```
html.Div([      html.Label("Select Statistics:"),      dcc.Dropdown(      i
d='.....',      options=.....,      value='..
.....',      placeholder='.....'      )
]),
```

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2. When encountering a Callback error updating output. container.children

Solution: Ensure **Input variables** used within the **callbacks** are defined and correctly scoped in the correct order.

```
@app.callback(Output(component_id='output-container', component_property='children'),
[Input(component_id='dropdown-
statistics', component_property='value'), Input(component_id='select-
year', component_property='value')])
```

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3. Graph not displaying

Solution: Ensure you have used the **elif** statement with the correct variable.

```
elif (input_year and selected_statistics=='.....') :          yearly_data = data[data
['Year'] == .....]
```



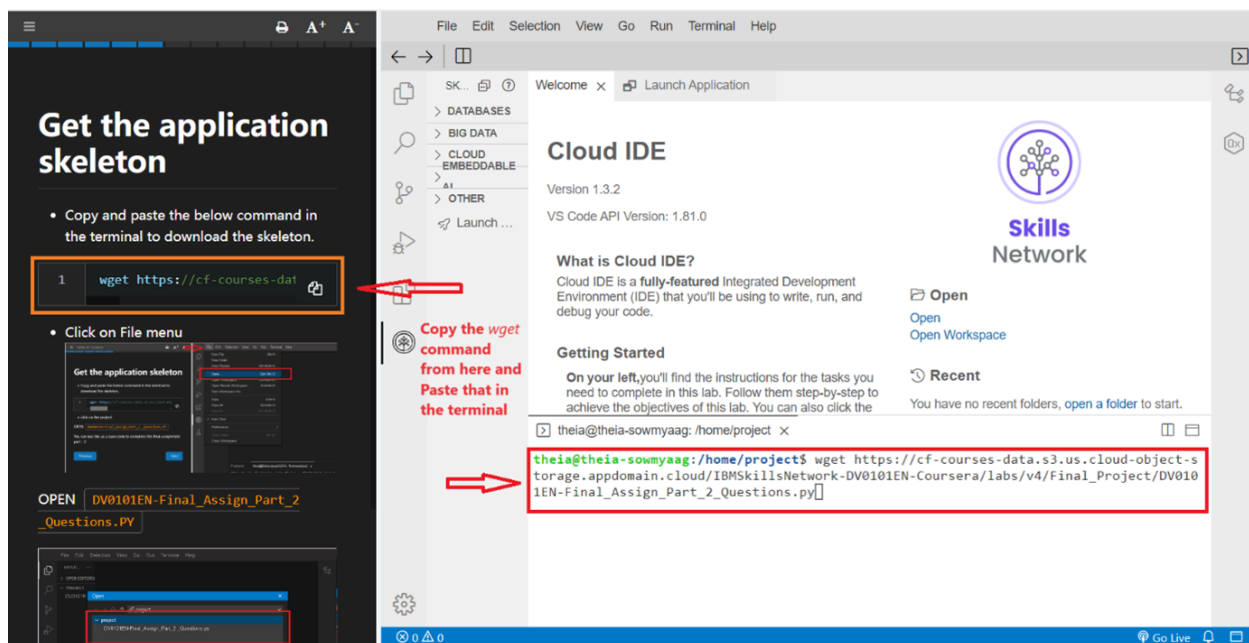
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4. The file could not be located under the project

Solution: Copy the **wget** command and paste it into the **terminal** to execute. The file will then be in the **Project directory**.



5. Port 8050 is in use by another program. Either identify and stop that program or start the server with a different port

Solution: Try with different port as shown below:

```
if __name__ == '__main__':app.run_server(port = 8090)
```



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